PROJECT EXPO 2026

Smart Hospital Care System

AI-Based Patient Monitoring and Predictive Alert System

TEAM MEMBERS:

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Smart Hospital Care System is an embedded healthcare monitoring solution designed to reduce manual monitoring workload and support faster nursing response. The system focuses on two key parameters: **IV fluid status** and **patient bed occupancy**. A load cell with HX711 is used to monitor the IV fluid weight and estimate the remaining fluid and time, while multiple FSR sensors detect patient presence and bed-vacancy events.

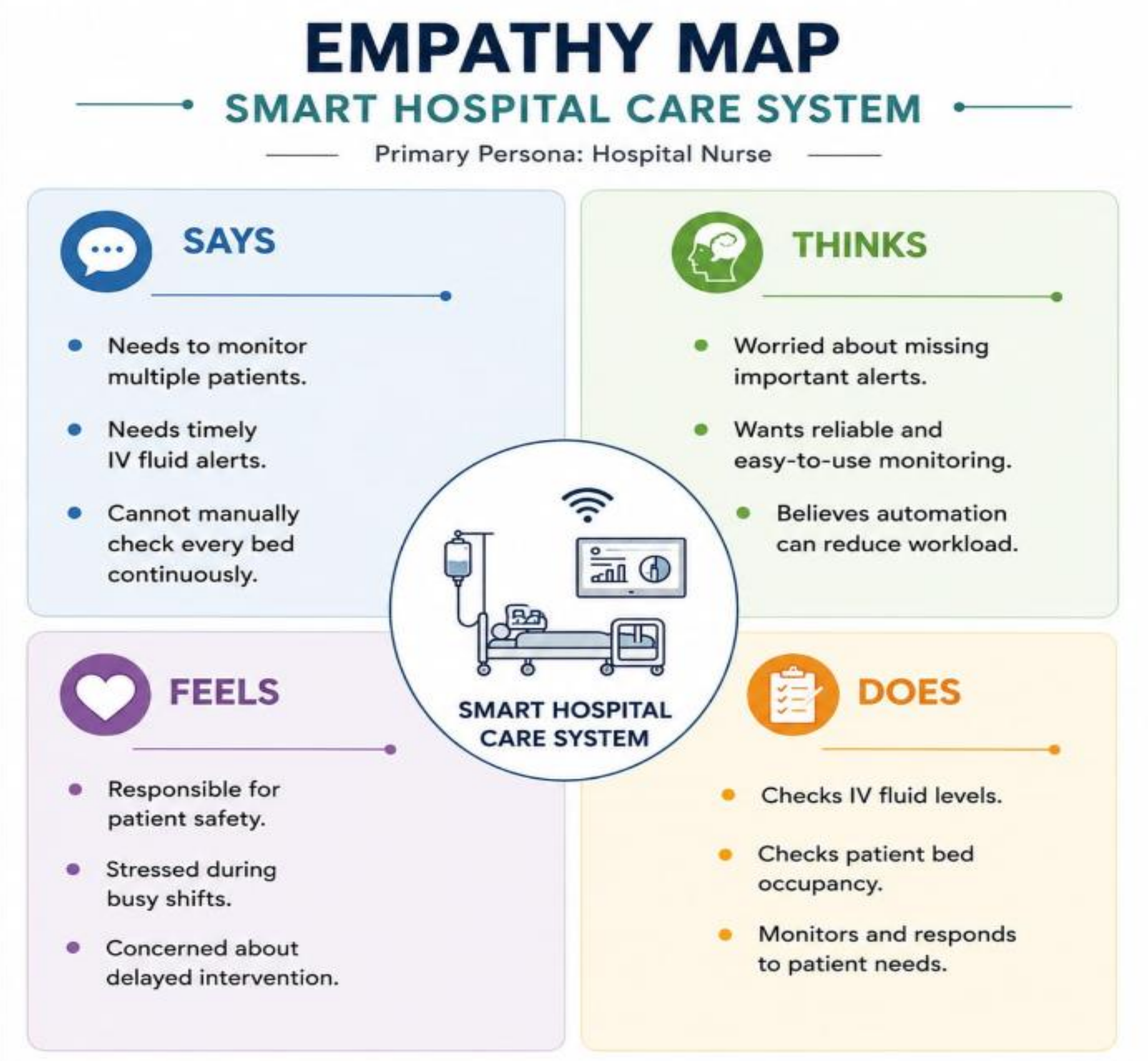
Brainstorming Report:

Idea Prioritization Matrix:

Idea	Feasibility	Innovation	Cost	User Benefit	Priority
IV Fluid Monitoring	High	High	Low	Very High	Very High
Bed Occupancy Monitoring	High	Medium	Low	High	High
Predictive IV Time	Medium	Very High	Low	Very High	Very High
Wireless Dashboard	High	High	Medium	High	High
Patient Attention Score	Medium	High	Low	High	High

Key Findings:

- IV fluid monitoring was selected as the primary feature because it directly improves patient safety.
- Bed occupancy monitoring enhances patient supervision with minimal hardware.
- Predictive IV time estimation provides proactive alerts instead of reactive notifications.
- Wireless monitoring reduces the need for frequent manual inspection.



User Persona:

Name : Priya Sharma
Age : 32 Years
Occupation : Staff Nurse
Experience : 6 Years

Responsibilities

- Monitor IV fluid levels.
- Observe patient bed occupancy.

- Respond to patient emergencies.
- Maintain nursing records.

Goals

- Reduce manual monitoring.
- Improve patient safety.
- Receive timely alerts.
- Save time during rounds.

Challenges

- Monitoring multiple patients simultaneously.
- Delayed identification of empty IV bottles.
- Increased workload during busy shifts.